

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103**

Summer Semester Examination - May - 2019

Course: B. Pharm.

Semester: IV

Subject with Subject Code: Pharmaceutical Organic Chemistry-III (BP401T)

Date: 27/05/2019

Marks: 75

Duration: 3hrs

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. 1 Choose the correct alternative.

20

1. A molecule is said to be chiral if-
 - A. It has four different atoms attached to carbon
 - B. Non super imposable on its mirror image
 - C. Superimposable on its mirror image
 - D. Both A & B
2. Specific rotation is denoted by-
 - A. R
 - B. S
 - C. D
 - D. $[\alpha]$
3. Angle strain is maximum in-
 - A. Cyclopropane
 - B. Cyclobutane
 - C. Cyclopentane
 - D. Cyclohexane
4. Racemic mixture can be separated by-
 - A. Mechanical Separation
 - B. Biochemical separation
 - C. Chemical Method
 - D. All of the above
5. Stereoisomers which are not mirror images of each other are called-
 - A. Enantiomers
 - B. Diastereomers
 - C. Racemic mixtures
 - D. None of the above
6. Pyridine is basic in nature-
 - A. Due to presence of lone pair of electrons on nitrogen atom
 - B. Due to formation of salts with acid
 - C. Due to formation of quaternary salt with alkyl halide
 - D. All of the above
7. Electrophilic substitution reactions occurs in Quinoline at-
 - A. Position 2 & 4
 - B. Position 3 & 5
 - C. Position 5 & 8
 - D. Position 4 & 6
8. Which of the ring in isoquinoline gets easily reduced?
 - A. Benzene ring
 - B. Nitrogen containing ring
 - C. Both rings
 - D. None of the above
9. In indole which of the heterocyclic ring is fused with benzene ring-
 - A. Pyrroazole
 - B. Imidazole
 - C. Isoxazole
 - D. Pyrrole

10. Sulphonation of indole in presence of pyridine occurs at-

- A. Position 2
- B. Position 1
- C. Position 3
- D. Position 4

11. In acridine which of the following catalyst causes reduction of benzene ring?

- A. Zn/HCl
- B. Pt/HCl
- C. LiAlH₄
- D. Both A & C

12. In thiazole and oxazole nucleophile and electrophile attacks at-

- A. C1 & C2 respectively
- B. C2 & C3 respectively
- C. C2 & C3 respectively
- D. C1 & C5 respectively

13. In pyrrole the number of electrons that the N atom contributes to the π -system is-

- A. 2
- B. 1
- C. 3
- D. 4

14. Electrophilic substitution in furan usually occurs at-

- A. C3 atom
- B. C2 atom
- C. Both C2 & C3 atoms
- D. None of the above

15. Which statement about thiophene is correct?

- A. Thiophene is polar
- B. Oxidative polymerization of thiophene leads to a conjugating polymer
- C. Thiophene is more reactive towards electrophiles than furan
- D. The S atom contributes two electrons to the π -system.

16. Which of the following is strong hydride donor?

- A. NaBH₄
- B. LiAlH₄
- C. B₂H₆
- D. None of the above.

17. Oppenaur oxidation is the reverse process of-

- A. Wolff-Kishner reduction
- B. Clemmensen reduction
- C. Meerwein-Ponndorf-Verley reduction
- D. Rosenmund's reduction

18. Birch reduction is done by using-

- A. Sodium in ammonia with alcohol
- B. Potassium in ammonia with alcohol
- C. Magnesium in ammonia with alcohol
- D. None of the above.

19. Beckmann's rearrangement is useful for synthesis of-

- A. Alcohols
- B. Amide
- C. Phenols
- D. Oxime

20. In Dakin reaction the -OH or -NH₂ group must be present at either which position of the reactant-

- A. Ortho position
- B. Meta position
- C. Para position
- D. Both A & C

Q. 2 Answer **any two** of the following questions.

20

- Give the methods of preparation, chemical reactions and medicinally important compounds for Pyrrole.
- Discuss Racemic Mixtures and their Resolution methods.
- Explain the reaction & mechanism of Wolf Kishner Reduction.

Q. 3 Answer **any seven** of the following questions.

35

- Give the Methods of preparation & chemical reactions for the Acridine.
- Discuss Methods of determination of configuration of Geometrical isomers.
- Give the Structure & medicinally important compounds for (Any-2)
 - Thiophene
 - Imidazole
 - Oxazole
 - Indole
- Write short note on Oppenaur Oxidation Reaction.
- Discuss the conformations of Butane.
- What is Claisen-Schmidt condensation? Give its mechanism & importance.
- Define Heterocyclic compounds & classify it.
- Explain relative aromaticity & reactivity of pyrrole & Thiophene.
- Explain the metal hydride reduction by using Sodium borohydride (NaBH_4).

End

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

WINTER SEMESTER EXAMINATION –DEC. 2019

Course. B. Pharmacy

Subject: Pharmaceutical organic chemistry III (BP401T)

Date: 04/12/2019

Semester: IV

Max Marks: 75

Time: 3hrs

Instructions:

- i. All questions are compulsory
- ii. Candidates should begin the answer to each question on new page.
- iii. Illustrate your answer with neat sketches, diagram, structures. Whichever necessary

Q. No1. Multiple choice Questions:

(20M)

(Each of the following questions has four alternatives. Only one of them is correct. Choose the correct answer.)

1. What is Stereoisomer?

- a) Isomers having a same molecular formula and same configuration
- b) Isomers having a same molecular formula but different configuration
- c) Isomers having a different molecular formula but the same configuration
- d) Isomers having a different molecular formula and different configuration

2. What are diastereomers?

- a) Molecules with non-superimposable mirror images
- b) Molecules with super imposable mirror images
- c) Molecules which do not have non-super imposable mirror images
- d) None of the above.

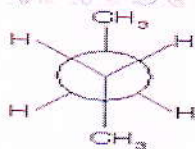
3. Which is following compounds will give cis-trans isomerism.

- a) 2-butene b) 2-butyne c) 2-butanol d) Butanal

4. Which of the isomer can be interconvert through rotation around a single bond.

- a) Conformers b) diastereomers c) enantiomers d) enantiomers

5. The conformation of n-Butane is



- a) Gauche
- c) Staggered

- b) Eclipsed
- d) None of the above

6. Isomers that have the same formula but differ in their structure are called

- a) Structural b) Optical c) Geometrical d) stereoisomer

7. Which of the following is NOT true about optical isomers?

- a) They involve carbon-carbon double bonds.
b) They are mirror images of each other.
c) They come in pairs.
d) They are not super imposable.

8. The molecular formula C_5H_{12} contains how many isomeric alkanes?

- a) 1 b) 2 c) 3 d) 4

9. Which among the following does not exhibit geometric isomerism

- a) 1-hexene b) 2-hexene c) 3-hexene d) 4-hexene

10. Compounds which have different arrangements of atoms in space while having same atoms bonded to each other are said to have

- a) position isomerism b) functional group isomerism
c) chain isomerism d) stereoisomerism

11. Which of the following is not a five-member ring?

- a) Pyridine b) Pyrrole c) Furan d) Thiophene

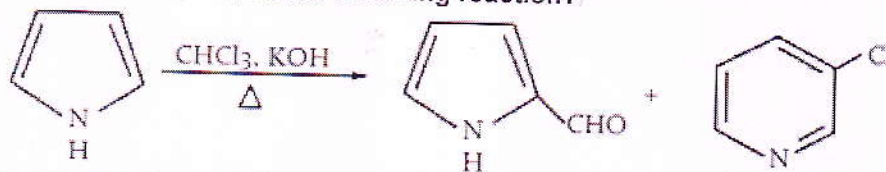
12. Which heterocyclic compound is more prone toward electrophilic substitution reaction.

- a) Pyrrole b) Furan c) Thiophene d) Pyridine

13. What is the product when thiophene reacts with Br_2 in benzene?

- a) 3,4-dibromothiophene b) 3-bromothiophene
c) 2,5-dibromothiophene d) 2-bromothiophene

14. What is the name of the following reaction?



- a) Gattermann reaction b) Riemeier tiemann reaction
c) Friedal craft reaction d) Blanc's chloromethylation

15. Which of the following order of reactivity is correct.

- a) furan > thiophene > pyrrole b) pyrrole > furan > thiophene

c) furan > pyrrole > thiophene

d) thiophene > pyrrole > furan

16. Which of the following statement is incorrect?

a) In imidazole, each N atom contributes one electron to the π -system.

b) Pyrazine is a diazine

c) Pyrimidine and pyrazine are isomers.

d) 4-Methylimidazole and 5-methylimidazole are tautomers.

17. Clemmensen reduction of ketone is carried out in presence of

a) conc H_2SO_4 .

b) Zinc amalgam (Zn/Hg) and Hydrochloric acid

c) $\text{K}_2\text{Cr}_2\text{O}_7$

d) HNO_3 and conc H_2SO_4

18. Pyridine undergoes nucleophilic substitution with NaNH_2 at 100°C to give which of the following?

a) 3-aminopyridine

b) 2-aminopyridine

c) 3,5-diaminopyridine

d) 2,5-diaminopyridine

19. Wolf Kishner reduction is-

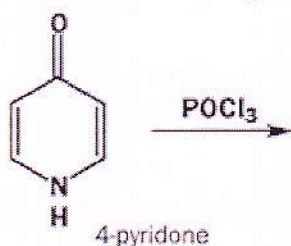
a) Reduction of aldehyde and Ketone

b) Alkane is obtained after reduction

c) in presence of hydrazine as reagent

d) all of the above.

20. What will be the product of the following reaction?



a) 3-chloropyridine

b) 2-dichloropyridine

c) 3,5-dichloropyridine

d) 2,5-dichloropyridine

Q.2. Attempt any two of the following:

(20M)

a) What are diastereomer and meso compounds? Are the meso compounds optically active or inactive? Explain with suitable example.

b) Describe synthesis, reaction, and medicinal uses of pyrrole. Add a note on aromaticity of pyrrole.

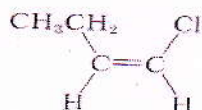
- c) Write in detailed reaction, mechanism, stereochemistry and application of the Birch reduction.

Q.3 Attempt any seven of the following:

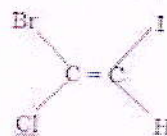
(35M)

- A. Classify isomerism in detailed. Explain structural isomerism with examples.
B. Assign the E and Z nomenclature of following organic compounds

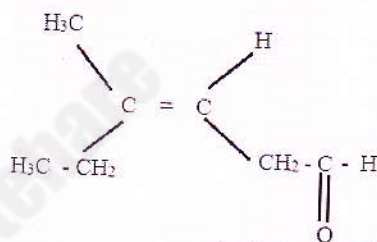
a)



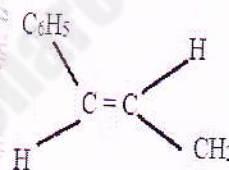
b)



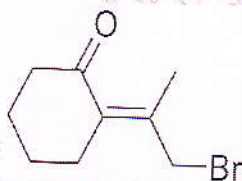
c)



d)



e)



C. Discuss electrophilic substitution reaction of thiophene. Why thiophene is more aromatic and more reactivity than pyrrole and furan.

D. What is reduction reaction? Describe reduction reaction of Quinoline and Isoquinoline.

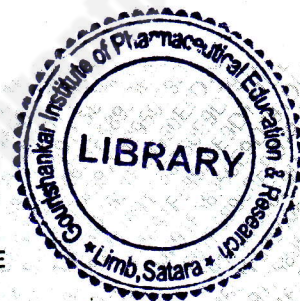
E. Write a note on Fischers Indole synthesis.

F. Draw the structure of a) Azepine, b) acridin, c) Imidazole, d) Oxazole, e) Pyrimidine.

G. Write methods of preparation of pyridine.

H. Describe various methods for the resolution of racemic mixture.

I. Write a note on conformation of n-Butane



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2020

Course: B. Pharmacy

Sem: IV

Subject Name: Pharmaceutical Organic Chemistry-III

Subject Code: BP401T

Max Marks: 75

Date: 24/08/2022

Duration: 3.45 Hrs.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

I. Multiple Choice Questions**(20 Marks)**

- 1) Which of the following prefix is used for Sulphur?
a) Oxa b) Aza c) Thia d) Sila
- 2) Number of optical active isomers in tartaric acid are
a) 2 b) 4 c) 6 d) 8
- 3) Furan react with Ammonia in the presence of aluminum trioxide at 400°C to give
a) Pyridine b) Furfural c) Pyrrole d) Furoic acid
- 4) Enantiomers are ---
a) Mirror images of each other b) Not mirror images of each other
c) Not concern with mirror images d) Diastereomers of each other
- 5) Suffix used for three membered heterocyclic saturated compound containing N is
a) Etidine b) Epidine c) Iridine d) Inidine
- 6) Notation used for dextro rotatory compounds is
a) D b) L c) d d) l
- 7) Electrophilic substitution in Thiophene usually occurs at
a) O atom b) C-2 Position c) C-3 Position d) Both A&B
- 8) Which of the following reagent reacts with pyrrole to form 2-formylpyrrole
a) HCOOH b) CHCl_3 & KOH c) H_2O_2 d) $(\text{CH}_3\text{CO})_2\text{O}$
- 9) Conversion of ketoximes to N-substituted amides takes place in ----- reaction
a) Beckmann b) Birch c) Claisen Schmidt d) Dakin



- 10) n-Propyl alcohol and isopropyl alcohol are examples of isomerism.
a) Functional b) Position c) Chain d) Metamerism
- 11) In -----synthesis 1,4- diketone is used as starting material
a) Fiest Benary b) Paal Knorr c) Traube d) Bischler
- 12) The synonym for pyrrole is _____.
a) Furan b) Pyrrole c) Pyridine d) Benzopyridine
- 13) Thiophene contains following heteroatom
a) S b) N c) O d) C
- 14) Following is the core component in structure of chloroquine
a) Isoquinoline b) Quinoline c) Thiophene d) Pyridine
- 15) Pyridine is ----- in nature
a) Weakly acidic b) Weakly basic c) Strong Acidic d) Neutral
- 16) What is the order of reactivity of pyrrole, furan & thiophene towards electrophile
a) furan, pyrrole, thiophene b) pyrrole, furan, thiophene
c) thiophene, pyrrole, furan d) furan, thiophene, pyrrole
- 17) Zinc amalgam & HCl is used as reagent in
a) Wolf-Kishner reduction b) Clemmensen reduction c) Birch reduction d) Dakin reaction
- 18) Wolff Kishner reduction mechanism begins with the formation of a
a) Hydride ion b) Hydrazone anion c) Nitrogen anion d) Nitrogen cation
- 19) ----- isomerism is observed in doubly bonded compounds
a) Structural b) Stereo c) Geometrical d) Optical
- 20) Identify the chiral molecule among the following
a) Isopropyl alcohol b) 2-pentanol c) 1-bromo 3-butene d) Isobutyl alcohol

II. Long Answers (Answer any Two)

(20 Marks)

- 1) Define and explain with examples the asymmetric synthesis.
- 2) Explain in detail conformations and conformational analysis of n-Butane and Cyclohexane.
- 3) Give principle, reaction and mechanism of Birch reduction and Wolf Kishner reduction.

III. Short Answers (Answer any Seven)

(35 Marks)

- 1) Define Biphenyl compounds and explain atropisomerism.
- 2) Explain the elements of symmetry with suitable examples.
- 3) Define Stereospecific reaction and explain it with examples.

- 4) Write the EAS reactions and medicinal uses of furan.
- 5) Write methods of preparation & reactions of Thiophene.
- 6) Give reaction mechanism of Paal-Knorr synthesis in pyrrole.
- 7) Why pyridine is much basic than pyrrole; explain on the basis of resonance.
- 8) Outline the method of synthesis of Indole.
- 9) Write methods of preparation & medicinal uses of pyrimidine.

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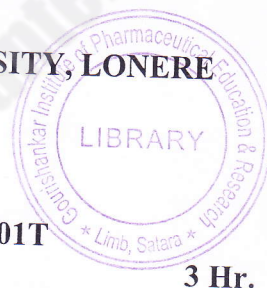
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 07/02/2023

Course: B. Pharmacy
Subject Name: Pharmaceutical Organic Chemistry-III
Max Marks: 75

Sem: IV
Subject Code: BP401T
Duration:



3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2)

- i) Give the structure and medicinal uses of Pyridine.
- ii) What is Dakin reaction?
- iii) Write the medicinal uses of Thiazole and Imidazole.
- iv) Give the structure of Thiophene and Oxazole.
- v) What are chiral and achiral molecules?
- vi) Write a note on Metal hydride reduction.
- vii) Define atropisomerism with example.
- viii) What is optical activity?
- ix) Give the structure and medicinal uses of Acridine.
- x) Define and explain Mesocompound.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10)

- i) Write the EAS reactions, synthesis and medicinal uses of Pyrrole and Furan.
- ii) Explain in detail about confirmations of n-Butane and Cyclohexane.
- iii) Define racemic modification. Describe the methods of resolution of racemic mixture.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) =

- i) Explain the terms with suitable example
a) Diastereoisomerism b) Enantiomerism
- ii) Outline the methods of synthesis of Indole.
- iii) Give the reaction and mechanism involved in Beckmann's rearrangement.
- iv) Write a note on basicity of pyridine.
- v) Describe the Birch reduction reaction.
- vi) Write a note on R and S system of configuration.
- vii) Give the reaction and mechanism of Oppenauer oxidation reaction.
- viii) Explain in detail asymmetric synthesis.
- ix) Give a brief account of stereospecific and stereoselective reactions.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2023

Date: 10/07/2023

Course: B. Pharmacy

Subject Name: Pharmaceutical Organic Chemistry-III

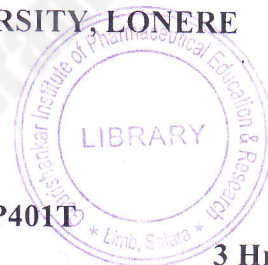
Max Marks: 75

Sem: IV

Subject Code: BP401T

Duration:

3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2)

- i) Give the structure and medicinal uses of Pyridine.
- ii) What is Dakin reaction?
- iii) Write the medicinal uses of Thiazole and Imidazole.
- iv) Give the structure of Thiophene and Oxazole.
- v) What are chiral and achiral molecules?
- vi) Write a note on Metal hydride reduction.
- vii) Define atropisomerism with example.
- viii) What is optical activity?
- ix) Give the structure and medicinal uses of Acridine.
- x) Define and explain Mesocompound.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10)

- i) Write the EAS reactions, synthesis and medicinal uses of Pyrrole and Furan.
- ii) Explain in detail about conformations of n-Butane and Cyclohexane.
- iii) Define racemic modification. Describe the methods of resolution of racemic mixture.

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(7 x 5) =

- i) Explain the terms with suitable example
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- v) Describe the Birch reduction reaction.
- vi) Write a note on R and S system of configuration.
- vii) Give the reaction and mechanism of Oppenauer oxidation reaction.
- viii) Explain in detail asymmetric synthesis.
- ix) Give a brief account of stereospecific and stereoselective reactions.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – Winter 2023

Date: 02/01/2024

Course : B. Pharmacy

Sem: IV

Subject Name : Pharmaceutical Organic Chemistry-III

Subject Code : BP401T

Max Marks : 75

Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define isomerism. Enlist its classification.
- ii) Calculate optical isomers of compound having 2 chiral carbon atoms.
- iii) Differentiate between stereospecific and stereoselective reaction.
- iv) Define cis and trans isomers with example.
- v) Draw the structure of furan. Give any two medicinal uses with example.
- vi) Write any one chemical reaction of thiophene.
- vii) Draw the structure and write any two medicinal uses of purine.
- viii) Write any one method of synthesis of Imidazole.
- ix) Define Oppenauer oxidation reaction with example.
- x) Define Clemmensen reduction reaction with general reaction.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Explain pyrrole in detail.
- ii) Discuss synthesis, reactions and medicinal uses of thiazole.
- iii) Explain the concept of optical isomer with its nomenclature systems.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Illustrate the concept of resolution of racemic mixture with its methods.
- ii) Write a note on atropisomerism.
- iii) Explain determination of configuration of geometrical isomer.
- iv) What is conformer? Explain conformational isomerism in Cycloalkane.
- v) Discuss relative aromaticity of pyrrole, furan and thiophene.
- vi) Outline the methods of synthesis of quinoline.
- vii) Comment on basicity of pyridine.
- viii) Explain principle, reaction and mechanism of Wolf Kishners reduction.
- ix) Give the reaction and mechanism of Metal hydride reduction.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: IV

Subject Name: Pharmaceutical Organic Chemistry-III

Subject Code: BP401T

Max Marks: 75

Date: 13-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1.	Objective Type Questions (Answer All the Questions)	(10 X 2) = 20
i)	Define optical activity with suitable example	
ii)	Give the Structure and medicinal uses of azepine	
iii)	Define Meso compound with suitable example	
iv)	Define Chiral and Achiral Molecule	
v)	How will you obtain 2-nitro pyrrole from pyrrole?	
vi)	Write down structure and medicinal uses of quinoline and imidazole	
vii)	Write Paul-Knorr synthesis for pyrrole.	
viii)	Define heterocyclic compound and classify them	
ix)	What is E Z geometrical isomerism? Give suitable example	
x)	What is Wolff-Kishner reduction reaction? Give general reaction	
Q.2.	Long Answers (Answer 2 out of 3)	(10 X 2) = 20
i)	Explain Stereospecific and Stereoselective reactions with suitable example	
ii)	What is racemic modification? Explain resolution of racemic mixture	
iii)	Define Conformation. Explain in detail about conformation of ethane and n-butane.	
Q.3.	Short Answers (Answer 7 out of 9)	(5 X 7) = 35
i)	Distinguish between enantiomerism and diastereoisomerism	
ii)	Explain in detail sequence rule with suitable example	
iii)	What is geometrical isomerism? Write method of determination of geometrical isomerism	
iv)	Write Skraup Quinoline synthesis with any two chemical reactions	
v)	Write the EAS reactions, synthesis and medicinal uses of Furan	
vi)	Give the reaction and mechanism involved in Schmidt reaction	
vii)	What is metal hydride reaction? Explain reaction and mechanism involved in NaBH_4	
viii)	Explain elements of symmetry	
ix)	Explain relative aromaticity and reactivity of Pyrrole, Furan and Thiophene	
*** END OF THE PAPER ***		

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